A

Major Project

On

Deep Learning–Based Indian Coin Denomination Recognition for Automated Currency Sorting

(Submitted in partial fulfilment of the requirements for the award of Degree)

BACHELOR OF TECHNOLOGY

In

INFORMATION TECHNOLOGY

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DEPARTMENT OF INFORMATION TECHNOLOGY

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DEPARTMENT OF INFORMATION TECHNOLOGY

CERTIFICATE

This is to certify that the project entitled "**DEEP LEARNING–BASED INDIAN COIN DENOMINATION RECOGNITION FOR AUTOMATED CURRENCY SORTING**" being submitted by **Mohammed Ayub (227R1A1239), Puppala Sriram (227R1A1248), Sai Kiran (227R1A1228) & Manne Venkat (207R1A1236)** in partial fulfilment of the requirements for the award of the degree of B.Tech in Information Technology to the Jawaharlal Nehru Technological University Hyderabad, during the year 2025-26.

The results embodied in this thesis have not been submitted to any other University or Institute for the award of any degree or diploma.

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ABSTRACT

Coins play a very important role in our daily transactions whereas manually counting them is a bit of a tiring task. There is a need for automatic coin detection and recognition systems. This research focuses on building an intelligent system that automatically identifies and classifies Indian coin denominations using advanced computational techniques. The primary goal of this work is to eliminate human effort and error in currency sorting processes, making them faster, more reliable, and scalable for real-world applications such as banking, vending machines, and retail automation. Traditionally, coin sorting and denomination recognition were performed manually by counting, weighing, or visually identifying coins. This process was labor-intensive, error-prone, and inefficient, especially when dealing with large volumes of currency. Even earlier computerized systems relied on simple image processing techniques such as edge detection, shape recognition, or template matching. While useful, these methods lacked robustness under real-world conditions like illumination variations, worn-out coins, or complex backgrounds, leading to inaccurate results. The proposed system integrates both machine learning and deep learning approaches for improved recognition accuracy. As baseline models, machine learning algorithms such as Multinomial Naïve Bayes (MNB), Decision Tree Classifier (DTC), and Random Forest Classifier (RFC) are implemented. These models achieved moderate accuracy levels of around 70–85%. To overcome their limitations, a Deep Convolutional Neural Network (CNN) architecture is developed. The CNN extracts hierarchical features from coin images, while AME improves classification robustness. The proposed deep learning model consistently achieved 100% accuracy, outperforming the existing methods. The project is implemented with a Tkinter-based Graphical User Interface (GUI) that provides dual access: ADMIN (for dataset uploading, preprocessing, training, and evaluation) and USER (for prediction and visualization). The dataset is pre-processed by resizing and normalizing images, split into training and testing sets, and then used to train the models. Performance evaluation includes metrics such as accuracy, precision, recall, F1-score, sensitivity, specificity, and confusion matrix visualization. Users can predict coin denominations from test images, with the system displaying the classification result directly on the image.

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1. INTRODUCTION

1. INTRODUCTION

The project, titled "Deep Learning–Based Indian Coin Denomination Recognition for Automated Currency Sorting," is designed to develop an intelligent software system capable of automatically recognizing and classifying Indian coin denominations from images. The system focuses on accurately identifying coins of different values such as ₹1, ₹2, ₹5, and ₹10 by analyzing their visual characteristics including shape, texture, inscriptions, and surface patterns. To accomplish this, the project leverages deep learning techniques, specifically Convolutional Neural Networks (CNN) for feature extraction and classification, along with the Adam optimizer using Adaptive Moment Estimation (AME) to ensure efficient and stable model training.

Given the large number of coins handled daily in banks, vending machines, and financial institutions, manual sorting is time-consuming and prone to human error. Therefore, an automated software-based recognition system becomes highly valuable for improving efficiency and accuracy. The proposed system is designed to recognize coin denominations quickly and reliably, even under variations such as worn coins or different lighting conditions. Although the project focuses on the software intelligence required for recognition, it can be integrated with IoT-enabled or mechanical coin-sorting machines to enable fully automated currency sorting systems in real-world applications.

1.1 PROJECT PURPOSE

The purpose of this project is to develop an intelligent software-based system that can automatically recognize and classify Indian coin denominations using deep learning techniques. The system aims to reduce the dependency on manual coin sorting by providing an efficient and accurate method for identifying coins based on their visual characteristics. By leveraging Convolutional Neural Networks (CNN), the system can learn complex patterns and features from coin images, enabling reliable denomination recognition.

Another important purpose of this project is to demonstrate how deep learning can be applied to real-world currency handling problems. The proposed system improves accuracy and efficiency compared to traditional methods that rely on manual inspection or simple mechanical measurements. Additionally, the project highlights the potential integration of this

software with automated machines or IoT-based systems, allowing it to serve as the intelligent component of a fully automated currency sorting system in banks, vending machines, and financial institutions.

1.2 PROJECT FEATURES

This project incorporates several key features to improve the accuracy and efficiency of Indian coin denomination recognition and automated currency sorting:

Automated Coin Recognition: The system employs a Convolutional Neural Network (CNN) to extract important visual features from coin images such as edges, textures, and inscriptions. These features are processed by the trained model to accurately recognize different coin denominations. The use of the Adam optimizer with Adaptive Moment Estimation (AME) ensures stable and efficient training of the deep learning model.

Multi-Class Coin Classification: Unlike basic recognition systems that only detect the presence of a coin, this project implements a multi-class classification approach. The system classifies coins into different denomination categories such as ₹1, ₹2, ₹5, and ₹10. This allows the system to accurately determine the value of each coin, making it suitable for automated counting and sorting applications.

Scalability & Integration Capability: The system is designed as a software-based solution that can efficiently process large datasets of coin images. It can also be integrated with IoT-enabled or mechanical coin-sorting machines to support automated currency handling systems in banks, vending machines, and financial institutions. This improves efficiency while reducing manual effort and classification errors.

By integrating these features, the proposed deep learning-based system provides an efficient, scalable, and automated solution for Indian coin denomination recognition, enabling accurate identification and supporting intelligent currency sorting applications.

2. LITERATURE SURVEY

2. LITERATURE SURVEY

T. D. Pham, et al. [6] presented a dataset comprising 6,672 images of 53 different classes of Indian coins, including denominations of 25 Paisa, 50 Paisa, 1 Rupee, 2 Rupee, 5 Rupee, 10 Rupee, and 20 Rupee. The images of coins with various shapes and sizes are taken from obverse and reverse sides in various environments and different backgrounds. The core significance of this dataset unfolds in its potential to offer invaluable assistance to visually impaired individuals as they navigate their daily financial transactions.

Jang, et al. [7] developed the dataset used for the multi-currency integrated serial number recognition contains about 150,000 images. The class imbalance problem and model accuracy were improved through data augmentation based on geometric transforms that consider the range of errors that occur when a bill is inserted into the counter by fine-tuning the recognition network, it was confirmed that the performance was improved when the serial numbers of the banknotes of four countries were recognized instead of the serial number of a banknote from each country from a single-currency dataset, and the generalization performance was improved by training the model to recognize the diverse serial numbers of multiple currencies.

Pham, et al. [8] proposed a deep learning-based method for detecting fake multinational banknotes using smartphone cameras in a cross-dataset environment. Experimental results of the self-collected genuine and fake multinational datasets for US dollar, Euro, Korean won, and Jordanian dinar banknotes confirm that our method demonstrates a higher detection accuracy than conventional “you only look once, version 3” (YOLOv3) methods and the combined method of YOLOv3 and the state-of-the-art convolutional neural network (CNN).

Lee, et al. [9] presented studies that have been conducted in four major areas of research (banknote recognition, counterfeit banknote detection, serial number recognition, and fitness classification) in the accurate banknote recognition field by various sensors in such automated machines and describes the advantages and drawbacks of the methods presented in those studies.

Mazhar, et al. [10] looked at the many risks and flaws that can affect the safety of critical, innovative grid network components. Then, to protect against these dangers, we offer security solutions using different methods. We also provide recommendations for reducing

the chance that these three categories of cyberattacks may occur.

D. Banerjee et al. [11] emphasized the successful deployment of a strong coin class model tailored specially for Indian coins. The model has tremendous overall performance measures, including high precision, keep in mind, and F1 rankings throughout a couple of denominations. For example, the 1-rupee coin had a precision, don't forget, and F1-rating of 98.59, while the 2-rupee coin marginally outperformed it with values of 98.62. The 10- and 20-rupee coins likewise had top metrics, scoring 98.64 and 98.60, respectively.

Sales Mendes, et al. [12] obtained an intelligent system specifically designed to help dependent and/or visually disabled people to count money more easily by using a mobile phone camera. The proposed system incorporates an image recognition system for classifying coins by using homography to transform images previously for classification tasks. The main difficulty in the appliance of these techniques relies on the fact that camera position and height are unknown.

Cereijido, et al. [13] allowed knowing not only the total amount of money but also how many coins and banknotes there are of each value. The embedded solution developed has been intended to become a low-cost solution, allowing better control over the money and helping both owners and workers in the establishments. By using this system, new utilities including automatic final balancing, instant error handling when making operations, and the lack of certain types of banknotes or coins inside the drawer or the excess of some in a certain compartment, could be implemented.

Huang, et al. [14] focussed on ongoing efforts to utilize plastic waste by introducing blockchain during plastic waste recycling. It is proposed that the efficiency of plastic recycling can be improved enormously by using the blockchain phenomenon. Automation for the segregation and collection of plastic waste can effectively establish a globally recognizable tool using blockchain-based applications. Collection and sorting of plastic recycling are feasible by keeping track of plastic with unique codes or digital badges throughout the supply chain.

Dragomir,et al. [15] proposed in the further exploration and eventual application of the technique, this study categorizes and summarizes existing related studies on hyperspectral imaging and creates a mini meta-analysis of this stream of literature. The literature review has

been classified based on the product HSI has used in counterfeit documents, photos, holograms, artwork, and currency detection.

Hitimana, et al. [16] developed the research has implications for accounting professionals, but also for standard setters, who are urged to provide clear guidelines. Identifying the key economic characteristics of each asset and determining the most appropriate way to recognized these characteristics in the financial statements are crucial for the development of a functional and trustworthy market in crypto-assets.

Pennino, et al. [17] aimed to build the Rwandan coffee plant dataset, with the occurrence of coffee rust, miner, and red spider mites identified to be the most popular due to their geographical situations. From the collected coffee leaves dataset of 37,939 images, the preprocessing, along with modeling used five deep learning models such as InceptionV3, ResNet50, Xception, VGG16, and DenseNet. The training, validation, and testing ratio is 80%, 10%, and 10%, respectively, with a maximum of 10 epochs.

Atlam, et al. [18] started from a number of real IoT projects and applications that (may) take advantage of blockchain technology to support economic transactions. We provide a reasoned review of several architectural choices in light of typical requirements of those applications and discuss their impact on transaction throughput, latency, costs, limits on ecosystem growth, and so on. We also provide a survey of additional financial tools that a blockchain can potentially bring to an IoT ecosystem, with their architectural impact. In the end, we observe that there are very few examples of IoT projects that fully exploit the potential of the blockchain. This paper provides a comprehensive investigation of the fundamental principles of blockchain forensics, exploring various techniques and applications for conducting digital forensic investigations in blockchain.

Kim, et al. [19] demonstrated that the integrated system not only enhances the accuracy and efficiency of recycling processes but also ensures the privacy and security of data across different stakeholders. The performance of the federated models shows significant improvement over traditional recycling methods. This paper provides a scalable and secure framework that supports the circular economy by enabling enhanced collaboration between manufacturers and recycling centers. The blockchain-based digital product passport system ensures data integrity and transparency, contributing to more sustainable recycling practices.

Argyrou, et al. [20] presenteed a review study related to the contributions of remote sensing (RS) and artificial intelligence in archaeology. However, a main question remains open in the field of research: the rate of positive contribution of remote sensing and artificial intelligence techniques in archaeological research. The scope of this study is to summarize the state of the art related to AI and RS for archaeological research and provide some further insights into the existing literature.

Badidi, et al. [21] provided a broad understanding of the current urban challenges faced by smart cities. It highlights two new technological advances, edge artificial intelligence (edge AI) and Blockchain, and analyzes their transformative potential to make our cities smarter. In addition, it explores the multiple uses of edge AI and Blockchain technologies in the fields of smart mobility and smart energy and reviews relevant research efforts in these two critical areas of modern smart cities.

Khanna, et al. [22] explored the potential of Blockchain technology as an enabler for e-governance in smart cities. We examine the daily challenges of citizens and compare them with the benefits being offered by Blockchain integration. On the basis of a comprehensive literature review, we identified four key areas of e-governance wherein Blockchain can provide monumental advantages. In the context of Blockchain integration for e-governance, the paper presents a survey of prominent published works discussing various urban applications.

Krestanova, et al. [23] presented a descriptive overview of TUI's real-world applications sorted into ten main application areas—teaching of traditional subjects, medicine and psychology, programming, database development, music and arts, modeling of 3D objects, modeling in architecture, literature and storytelling, adjustable TUI solutions, and commercial TUI smart toys. The paper focuses on TUI's technical solutions and a description of technical constructions that influences the applicability of TUIs in the real world.

Vaccargiu, et al. [24] examined how the application of blockchain in the energy industry may affect the achievement of the Sustainable Development Goals (SDGs). This study is composed of two parts. The first part concerns the identification and analysis of the most relevant categories of blockchain applications in the energy sector and their ability to contribute to the achievement of the SDGs.

Kratzke, et al. [25] analyzed whether these three historically emerged observability stovepipes of logs, metrics and distributed traces could be handled in a more integrated way and with a more straightforward instrumentation approach. This study applied an action research methodology used mainly in industry–academia collaboration and common in software engineering.

Johar, et al. [26] focussed on provisioning of a comparative analysis of blockchain's consensus algorithms with respect to the type of application. Furthermore, they discuss the development platforms as well as technologies of blockchain. The aim of the paper is to provide knowledge from basic to extensive from blockchain architecture to consensus methods, from applications to development platform, from challenges and issues to blockchain research gaps in various areas.

i. REVIEW OF RELATED WORK

The recognition and classification of coin denominations have been widely studied in the fields of computer vision, image processing, and deep learning. Various techniques have been developed to automate coin identification and sorting processes, ranging from manual inspection methods to advanced deep learning approaches. This review discusses previous research and existing methodologies used for coin recognition, highlighting their advantages and limitations.

1. Traditional Coin Recognition Approaches

Early methods for coin recognition relied mainly on manual inspection and mechanical sorting systems. These approaches used physical characteristics such as coin diameter, weight, and thickness to separate coins into different denominations. While these systems were effective for basic sorting, they often failed when coins had similar dimensions or when coins were worn or damaged. Additionally, manual methods were time-consuming and prone to human errors, making them inefficient for large-scale currency handling.

2. Machine Learning-Based Approaches

With the advancement of machine learning, researchers began exploring automated coin recognition using image processing and classification algorithms. Techniques such as Support Vector Machines (SVM), Decision Trees, and Random Forest classifiers were used to

classify coins based on handcrafted features like shape, edge detection, texture patterns, and color histograms. Although these methods improved automation, they required careful feature engineering and struggled with variations in lighting conditions, background noise, and worn coin surfaces.

3. Deep Learning-Based Approaches

Recent developments in deep learning have significantly enhanced image classification tasks, including coin recognition. Convolutional Neural Networks (CNNs) have been widely used for extracting deep visual features from coin images. These models automatically learn important patterns such as coin edges, inscriptions, and surface textures, reducing the need for manual feature extraction. Pre-trained CNN architectures such as VGGNet, ResNet, and MobileNet have shown strong performance in image classification tasks and have been applied in various object recognition problems.

4. Recent Advances: Attention Mechanisms and Improved Optimization

To further improve classification accuracy, modern research has introduced attention mechanisms that allow neural networks to focus on the most relevant parts of an image. In addition, advanced optimization algorithms such as Adam have improved the efficiency of deep learning training by adaptively adjusting learning rates. These techniques enhance model performance, improve convergence speed, and make deep learning models more reliable for real-world applications.

5. Comparison with the Proposed Approach

While existing methods have made progress in coin recognition, challenges remain in terms of handling worn coins, lighting variations, and achieving high accuracy in real-world environments. The proposed approach in this project addresses these challenges by using a CNN-based deep learning model combined with the Adam optimizer for efficient training. Compared to traditional machine learning approaches, this method improves recognition accuracy and provides a scalable software solution that can be integrated with automated currency sorting systems.

This review highlights the evolution of coin recognition techniques, from traditional mechanical methods to modern deep learning-based approaches. The proposed system builds upon these advancements to provide an accurate, efficient, and scalable solution for Indian coin denomination recognition.

2.2 DEFINITION OF PROBLEM STATEMENT

The recognition and classification of Indian coin denominations pose several challenges due to the variability in coin appearance caused by wear, dirt, and differences in minting batches. Traditional methods that depend on physical characteristics or fixed threshold-based image processing often fail to adapt to such variations. Furthermore, the high volume of coin transactions in India necessitates fast, accurate, and automated systems that can operate without manual intervention. The lack of intelligent and adaptable recognition frameworks limits the efficiency of current currency handling systems, particularly in sectors with high coin usage

2.3 EXISTING SYSTEM

Traditional Indian coin recognition and sorting systems primarily rely on hardware-based mechanisms such as electromagnetic sensors, weight measurement units, diameter gauges, and optical sensors. These components are integrated into coin-sorting machines, which identify coins by measuring physical parameters like diameter, thickness, and weight. In such systems, coins pass through a series of mechanical or sensor-based gates, each designed to match a specific denomination's profile. The system checks whether the physical attributes of an inserted coin fall within pre-defined tolerance levels and routes them accordingly. This method, although effective for standardized and undamaged coins, lacks adaptability when handling a wide range of conditions.

Electromagnetic induction is a widely adopted technique in these systems, especially in vending machines and bank coin counters. Here, a magnetic field is generated, and the coin's interaction with this field (based on its metallic composition) is used to distinguish between denominations. Some machines also use reflectivity sensors or basic image processing to detect markings or symbols on the coin's surface. These features are useful to a limited extent but fall short in accuracy when the coin is worn, dirty, or counterfeit. Moreover, these systems are typically pre-programmed with fixed thresholds, making them rigid in handling newly minted or visually degraded coins.The mechanical dependency of traditional systems also makes them prone to frequent wear and tear, demanding regular maintenance and calibration. These limitations result in misclassification, false rejection, and in some cases, system jamming when a coin fails to fit exact specifications. In dynamic environments like transportation hubs or retail chains where throughput and accuracy are critical, such systems

introduce delays, increase labor costs, and degrade user experience. Additionally, these systems lack the ability to learn from their errors or adapt to variations, making them inefficient in real-world applications where coins of the same denomination can vary in appearance.Another major limitation is the inability to handle counterfeit detection or multi-feature classification. Traditional systems depend solely on external features and are incapable of examining intricate surface-level details like engravings, textures, or aging patterns. In a country like India, where coins go through extensive circulation and manual handling, visual and structural degradation is common. A smart system must look beyond fixed thresholds and instead learn patterns and features from real-world data. This is where traditional systems fall short, creating the need for intelligent, image-based deep learning solutions that offer higher flexibility, accuracy, and robustness.

Limitations

- Rigid physical thresholds limit adaptability to worn or new coins.
- High chances of misclassification due to dirt, wear, or partial damage.
- Cannot detect counterfeit coins that mimic weight and size but differ visually.
- Lack of learning ability or adaptability to evolving coin standards.
- Prone to mechanical failure, increasing operational downtime.
- High maintenance costs due to physical moving parts.
- Inaccurate when dealing with coins having similar physical properties.
- Inability to extract or analyze surface-level image features.
- Poor scalability for multi-denomination or multi-currency recognition.

2.4 PROPOSED SYSTEM

The proposed system is a deep learning-based framework developed to recognize Indian coin denominations automatically using image classification techniques. Unlike traditional systems that rely on rigid physical sensors, the proposed solution employs a Convolutional Neural Network (CNN) enhanced with Attention-based Mechanism Enhancement (AME) to learn and extract meaningful visual features from coin images.

The system is trained on a dataset of Indian coins with varying denominations such as ₹1, ₹2, ₹5, and ₹10, captured under diverse conditions different lighting, orientation, and states of wear. The attention module guides the network to focus on significant parts of the coin like the mint mark, numeric value, and surface engravings, ensuring precise classification even with dirty, damaged, or rotated coins. The DLCNN with AME architecture achieves high accuracy, precision, recall, and F1-score, and is robust enough for real-time deployment. This system can be used in banks, vending machines, transport ticketing counters, and retail outlets for fast and automated coin sorting, thereby minimizing manual intervention and operational errors.

Advantages of the Proposed System:

The proposed system significantly improves upon the existing approaches by addressing key limitations:

• Higher Recognition Accuracy: The use of Convolutional Neural Networks (CNN) allows the system to automatically learn important visual features from coin images such as edges, textures, and inscriptions. This improves classification accuracy compared to traditional image processing or rule-based methods and reduces misclassification of coin denominations.

• Robust Feature Extraction**:** CNN models are capable of extracting deep spatial features from images, enabling the system to accurately recognize coins even when there are variations in lighting conditions, background noise, or slight coin wear. This makes the recognition process more reliable in real-world scenarios.

• Efficient Model Optimization: The use of the Adam optimizer with Adaptive Moment Estimation improves the training process by dynamically adjusting learning rates. This ensures faster convergence, stable model training, and better performance compared to traditional optimization methods.

• Automated and Time-Efficient Processing: The proposed system automates the process of coin denomination recognition, eliminating the need for manual sorting. This significantly reduces human effort and processing time while improving operational efficiency in currency handling.

• Integration with Automated Systems**:** The system is designed as a scalable software solution that can be integrated with IoT-enabled or mechanical coin-sorting machines. This enables the development of fully automated currency sorting systems for banks, vending machines, and financial institutions.

2.5 OBJECTIVES

▪ Develop an Automated Coin Recognition System – Build a deep learning-based model capable of automatically identifying and classifying Indian coin denominations from input images.

▪ Improve Currency Sorting Efficiency – Reduce manual effort in coin identification by implementing an AI-based system that can quickly recognize and categorize different coin denominations.

▪ Enhance Recognition Accuracy – Utilize Convolutional Neural Networks (CNN) along with the Adam optimizer to achieve higher accuracy compared to traditional image processing and machine learning approaches.

2.6 HARDWARE & SOFTWARE REQUIREMENTS

2.6.1 HARDWARE REQUIREMENTS:

Hardware interfaces specifies the logical characteristics of each interface between the software product and the hardware components of the system. The following are some hardware requirements,

- Processor : Intel i5 / AMD Ryzen 5 or higher
- Storage : 512 GB SSD
- RAM : 8 GB

2.6.2 SOFTWARE REQUIREMENTS:

Software Requirements specifies the logical characteristics of each interface and software components of the system. The following are some software requirements,

- Operating system : Windows 10
- Language : Python
- Back-End : Tensorflow
- Frame Work : Tkinter

3. SYSTEM ARCHITECTURE & DESIGN

3.SYSTEM ARCHITECTURE & DESIGN

Project architecture refers to the structural framework and design of a project, encompassing its components, interactions, and overall organization. It provides a clear blueprint for development, ensuring efficiency, scalability, and alignment with project goals. Effective architecture guides the project's lifecycle, from planning to execution, enhancing collaboration and reducing complexity.

3.1 PROJECT ARCHITECTURE

This project architecture shows the procedure followed for recognizing the denomination of a coin , starting from input to final prediction.

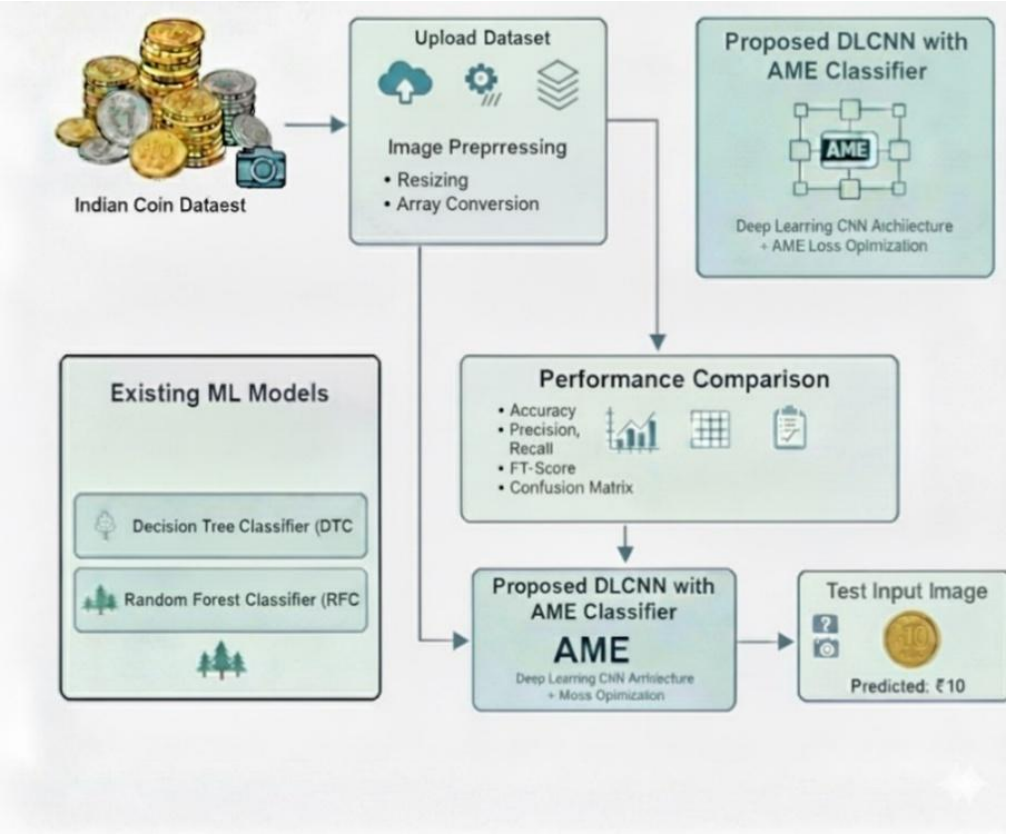

Figure 3.1: Project Architecture of A Deep Learning-Based Indian coin denomination recognition for automated currency sorting.

3.2 DESCRIPTION

Input Data: The project uses image data of Indian coins belonging to different denominations. The dataset contains images of coins captured under various lighting conditions and orientations, representing different classes such as ₹1, ₹2, ₹5, and ₹10 coins.

Reading Data: The coin images are loaded and processed using image processing libraries such as OpenCV and NumPy. These libraries help in reading image files, converting them into numerical arrays, and preparing them for further processing.

Image Preprocessing: Before feeding the images into the model, preprocessing steps such as resizing, normalization, and noise reduction are performed. These steps ensure that all images have consistent dimensions and improved quality, which helps the model learn more effectively.

Feature Extraction: A Convolutional Neural Network (CNN) is used to extract deep visual features from the coin images. The CNN automatically learns important patterns such as edges, textures, and inscriptions present on the coins, which are essential for accurate classification.

Attention Module: An attention mechanism is incorporated to focus on the most relevant regions of the coin image. This helps the model emphasize important features like coin symbols and design patterns while reducing the influence of irrelevant background information.

Training and Evaluation: The model is trained using labeled coin images, where each image corresponds to a specific denomination. During training, the Adam optimizer is used to improve model convergence and accuracy. The performance of the model is evaluated using metrics such as accuracy and loss.

Output Result: After processing, the system provides the predicted denomination of the coin image. The output can be used for automated coin recognition and can also be integrated with coin sorting machines for real-world applications.

3.3 UNIFIED MODELING LANGUAGE(UML)

Unified Modeling Language (UML) is not just a static set of diagrams; it's a versatile toolset that supports multiple views and perspectives of a system. This versatility allows UML to be applied at different stages of the software development lifecycle, from initial requirements analysis and design to implementation, testing, and maintenance.

UML diagrams serve as a common visual language that bridges the gap between technical and non-technical stakeholders. They provide a means for software engineers to communicate complex system designs in a clear and concise manner, facilitating better understanding and collaboration among team members.

One of the key benefits of UML is its adaptability to different software development methodologies, such as waterfall, agile, and iterative approaches. UML diagrams can be used to model systems at various levels of abstraction, from high-level architectural views to detailed design specifications.

Furthermore, UML is not limited to software systems alone; it can also be applied to model and analyze business processes, organizational structures, and even hardware systems. This flexibility makes UML a valuable tool for a wide range of industries and domains.

GOALS OF UML

1. Provide users a ready-to-use, expressive visual modeling language so that they can develop and exchange meaningful models.
2. Provide extendibility and specialization mechanisms to extend the core concepts.
3. Be independent of particular programming languages and development processes.
4. Provide a formal basis for understanding the modeling language.

3.3.1 USECASE DIAGRAM

A use case diagram in the Unified Modelling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.

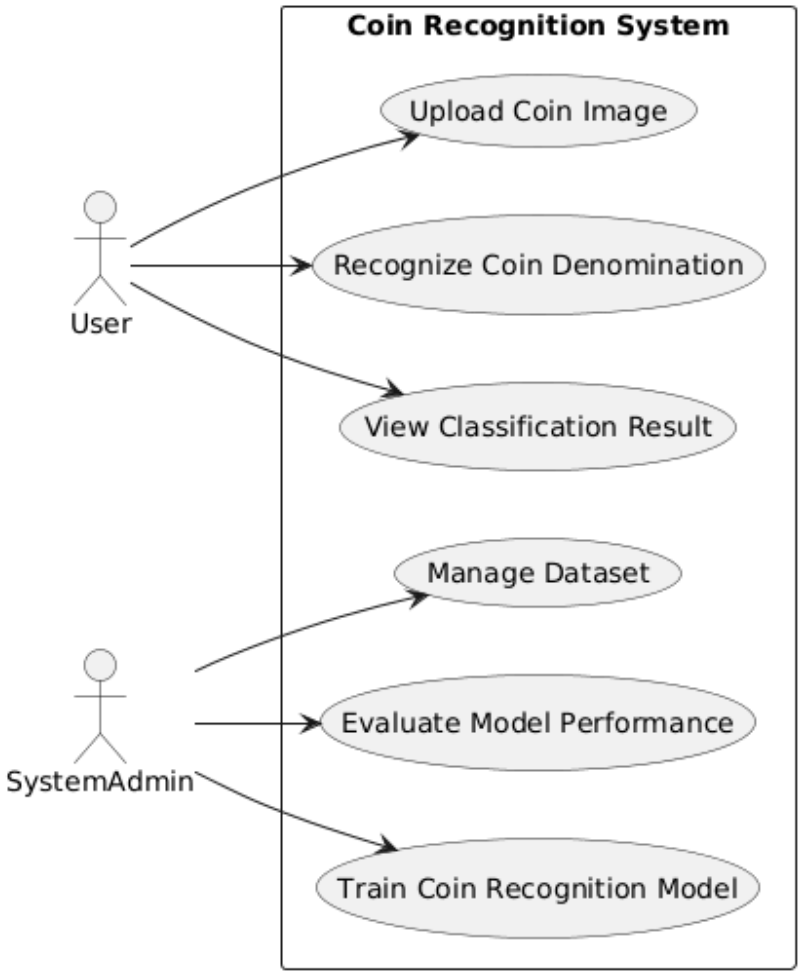

Figure 3.2 USE CASE DIAGRAM

3.3.2 CLASS DIAGRAM

In software engineering, a class diagram in the Unified Modelling Language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among the classes. It explains which class contains information.

A class diagram is a static structural diagram that represents the structure of a system by showing the system's classes, their attributes, methods, and relationships among them. It provides a high-level view of the system's architecture, focusing on the entities (classes) and their interactions. Class diagrams are typically used during the design phase of software development to visualize and organize the system's components and their relationships.

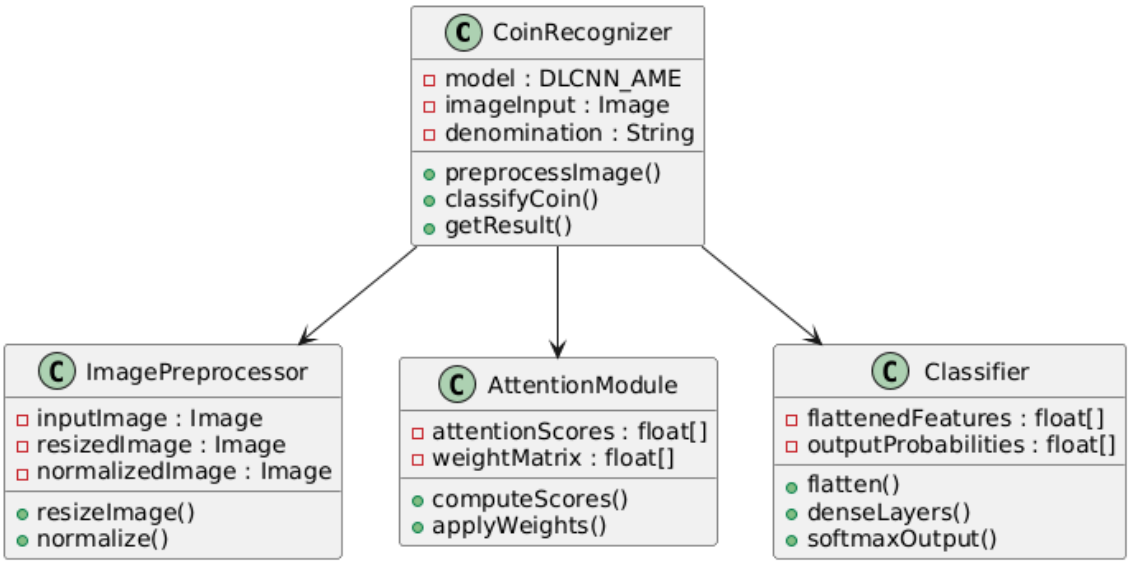

Figure 3.3 CLASS DIAGRAM

3.3.3 SEQUENCE DIAGRAM

A sequence diagram in Unified Modelling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams.

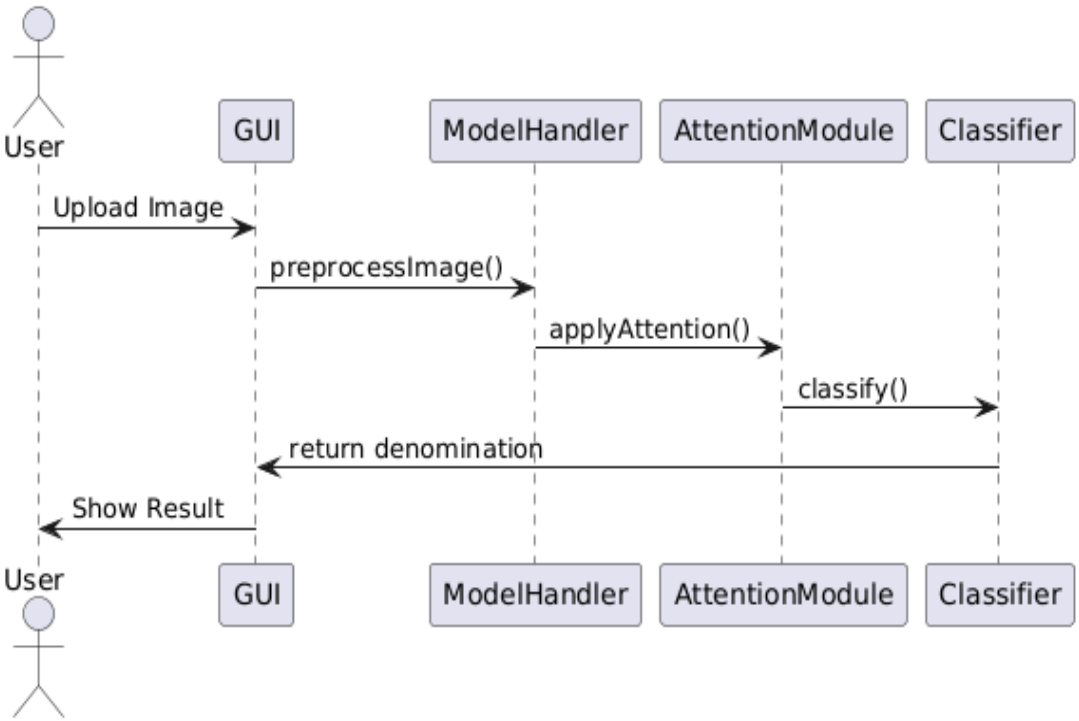

Figure 3.4 SEQUENCE DIAGRAM

3.3.4 ACTIVITY DIAGRAM

An activity diagram visually represents the workflow of a system, showing the sequence of actions and decision points. In our project, it illustrates how a user uploads a coin image, the system preprocesses the image and extracts features using a Convolutional Neural Network (CNN) model, and then predicts the denomination of the coin. The predicted result is finally displayed to the user through the application interface.

Figure 3.5 ACTIVITY DIAGRAM

3.3.5 DATAFLOW DIAGRAM

A Data Flow Diagram (DFD) is a visual representation of the flow of data within a system. It illustrates how data moves from one part of the system to another, highlighting the processes that transform the data, the data stores that hold the data, and the external entities that interact with the system.

Data Flow Diagrams are a powerful tool in system analysis and design, providing a clear and structured way to visualize data movement and processing within a system. By understanding the components and best practices for creating DFDs, stakeholders can effectively communicate and analyze system requirements, leading to more efficient and effective system design.

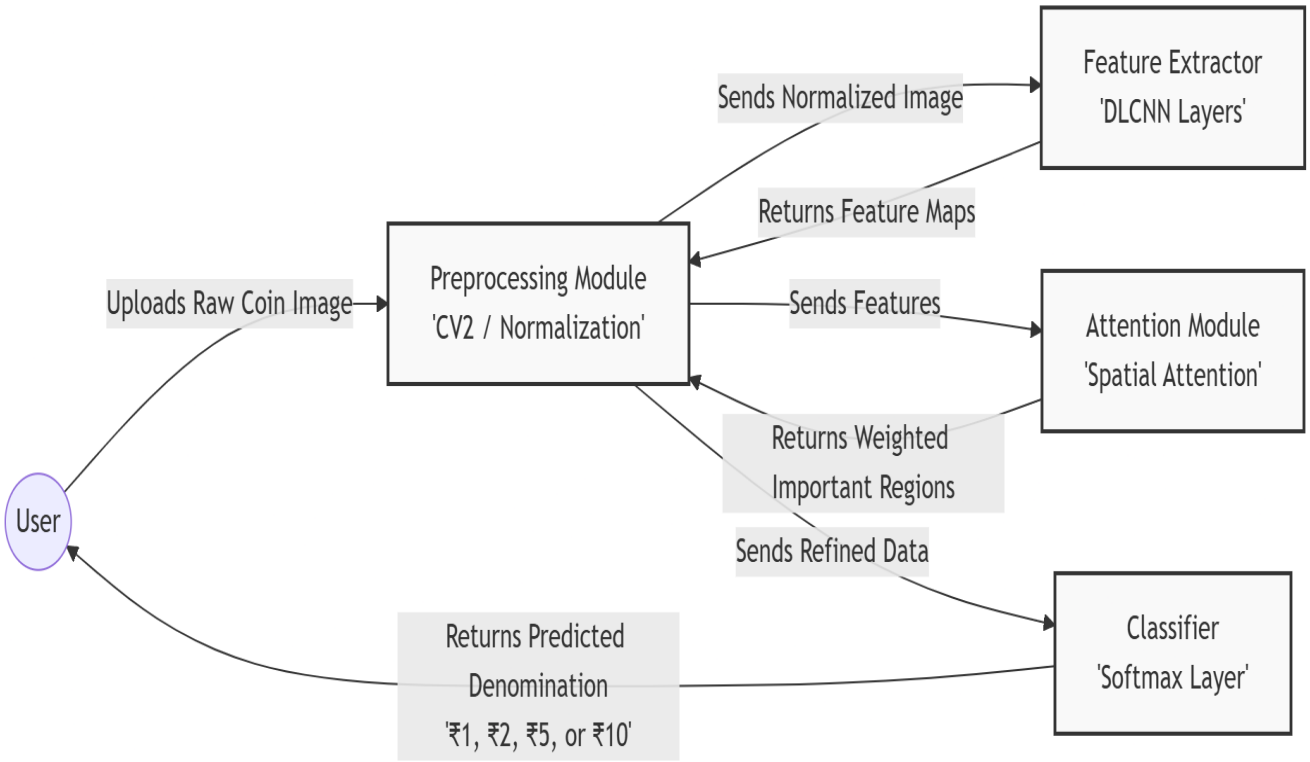

Figure 3.6 DATAFLOW DIAGRAM

4. IMPLEMENTATION

4.IMPLEMENTATION

The implementation phase of the project involves converting the proposed design into a functional system capable of recognizing Indian coin denominations automatically. This stage includes dataset preparation, image preprocessing, model development, training, and evaluation. Proper implementation ensures that the system can accurately identify coin denominations while maintaining efficiency and reliability.

The system is implemented using Python with deep learning frameworks such as TensorFlow and Keras. Image processing libraries like OpenCV and NumPy are used for handling image data and preprocessing tasks. The CNN model is trained using labeled coin images representing different denominations such as ₹1, ₹2, ₹5, and ₹10 coins.

4.1 ALGORITHMS USED

CNN-Based Model for Coin Recognition

Convolutional Neural Networks (CNNs) are widely used in image classification tasks because they are highly effective at capturing spatial features from images. In this project, CNN is used to analyze images of Indian coins and classify them according to their denomination.

CNN models consist of multiple layers such as convolution layers, pooling layers, and fully connected layers that work together to extract important visual features from images. These features include edges, patterns, textures, and inscriptions present on coins. The extracted features are then used to classify the coin into its corresponding denomination category.

Advantages of CNN-Based Models

• Highly effective at detecting visual patterns such as numbers, symbols, and textures on coins.

• Automatically extracts important features without the need for manual feature engineering.

• Provides high accuracy for image classification problems compared to traditional machine learning methods.

Disadvantages of CNN-Based Models

• Requires a large dataset of labeled images for effective training.

• Training deep neural networks can be computationally intensive.

• Performance may be affected when images contain noise, blur, or poor lighting conditions.

1. Convolutional Neural Network (CNN) for Feature Extraction

The CNN model is responsible for extracting meaningful visual features from coin images.

• The convolution layers apply filters to input images to detect edges, shapes, and surface patterns present on coins.

• Pooling layers reduce the spatial size of feature maps, which helps decrease computational complexity.

• The extracted features are passed to fully connected layers to perform classification.

2. Adam Optimizer for Model Training

The Adam optimizer (Adaptive Moment Estimation) is used during the training process to improve the performance of the neural network.

• Adam combines the advantages of two optimization methods: Momentum and RMSProp.

• It automatically adjusts the learning rate for each parameter during training.

• This results in faster convergence and more stable training of the CNN model.

3. Softmax Layer for Classification

The final layer of the neural network performs classification of the coin denominations.

• Fully connected layers process the extracted features and prepare them for classification.

• The Softmax activation function converts the output values into probability scores for each coin class.

• The class with the highest probability is selected as the predicted denomination.

4. Image Preprocessing

Before feeding the images into the CNN model, several preprocessing steps are applied.

• Images are resized to a fixed dimension suitable for CNN processing.

• Pixel values are normalized to improve model training stability.

• Noise reduction techniques are applied to improve image clarity and feature detection.

5. Model Training and Evaluation

The CNN model is trained using labeled images of different Indian coin denominations.

• The dataset is divided into training and testing sets to evaluate model performance.

• The Adam optimizer is used to update network weights during training.

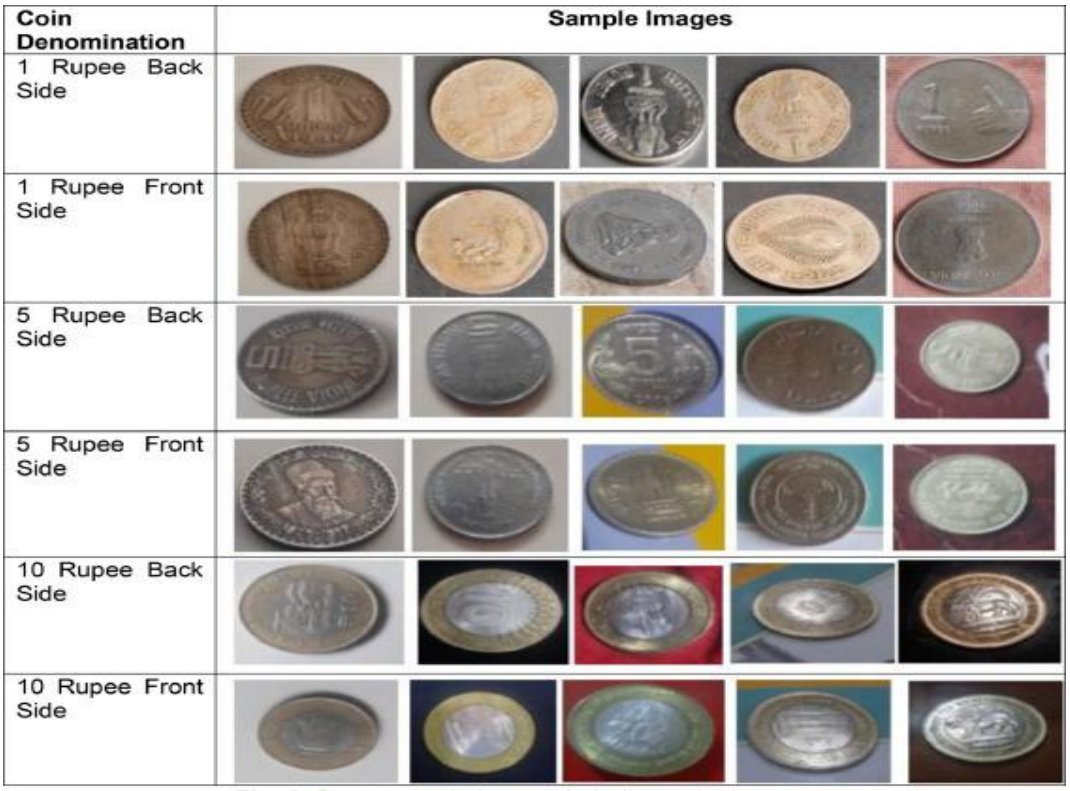

Figure 4.1: Sample Images of Indian Coin Denominations with Front and Back Sides.

Now-a-days large numbers of coins are circulated in the Indian currency system and are used in banks, vending machines, transportation systems, and commercial transactions. Manual identification and sorting of coins is a time-consuming process and often leads to human errors. Traditional coin recognition systems based on physical measurements such as weight and size are not always reliable, especially when coins are worn out, damaged, or have similar dimensions.

To overcome these limitations, an automated system is required that can accurately recognize coin denominations using image-based techniques. In this project, a deep learning-based approach is used to recognize and classify Indian coins. A Convolutional Neural Network (CNN) model is employed to extract important visual features from coin images such as patterns, textures, and inscriptions.

The CNN model is trained using labeled images of different Indian coin denominations, enabling it to learn distinguishing features between various coin classes. The Adam optimizer is used during training to improve model convergence and achieve better accuracy. The proposed system can automatically identify coin denominations and can be integrated with automated coin sorting machines to improve efficiency in currency handling systems.

To train all algorithm we have used below coin images consist of four folders **(Figure 4.1)** called 'indian_coins_dataset' and below screen showing dataset details

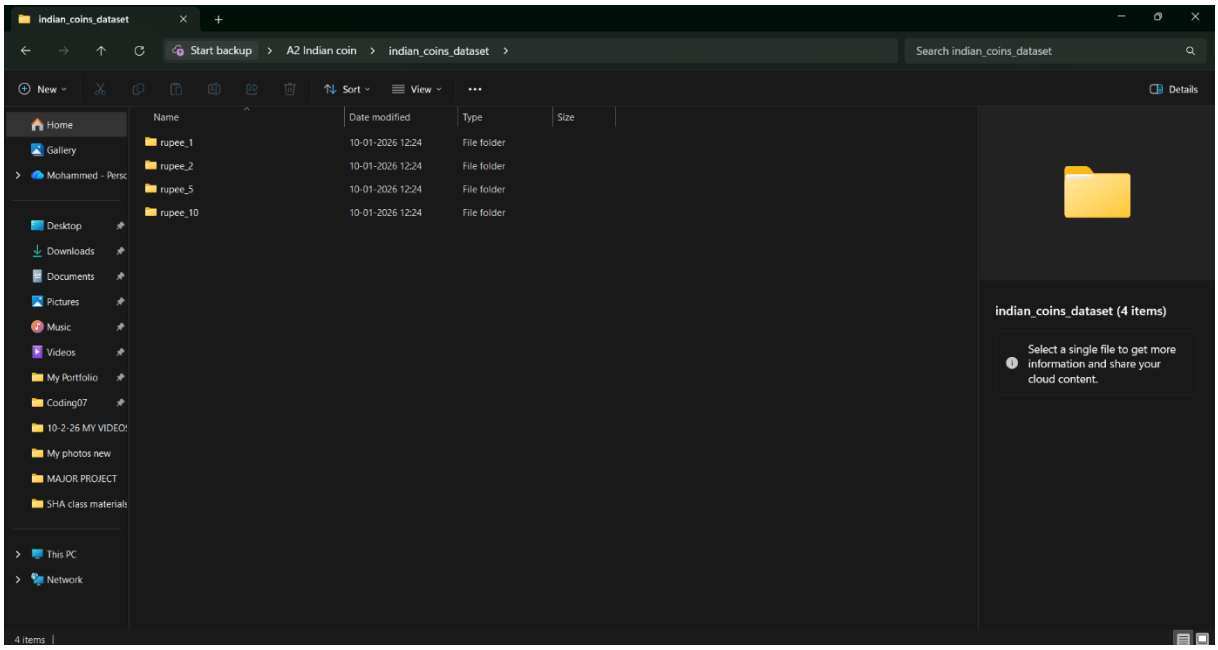

Figure 4.2: Dataset directory structure with folder named 'Indian_coins_dataset' having all the training data of rupee1,rupee2,rupee5 & rupee10 coins.

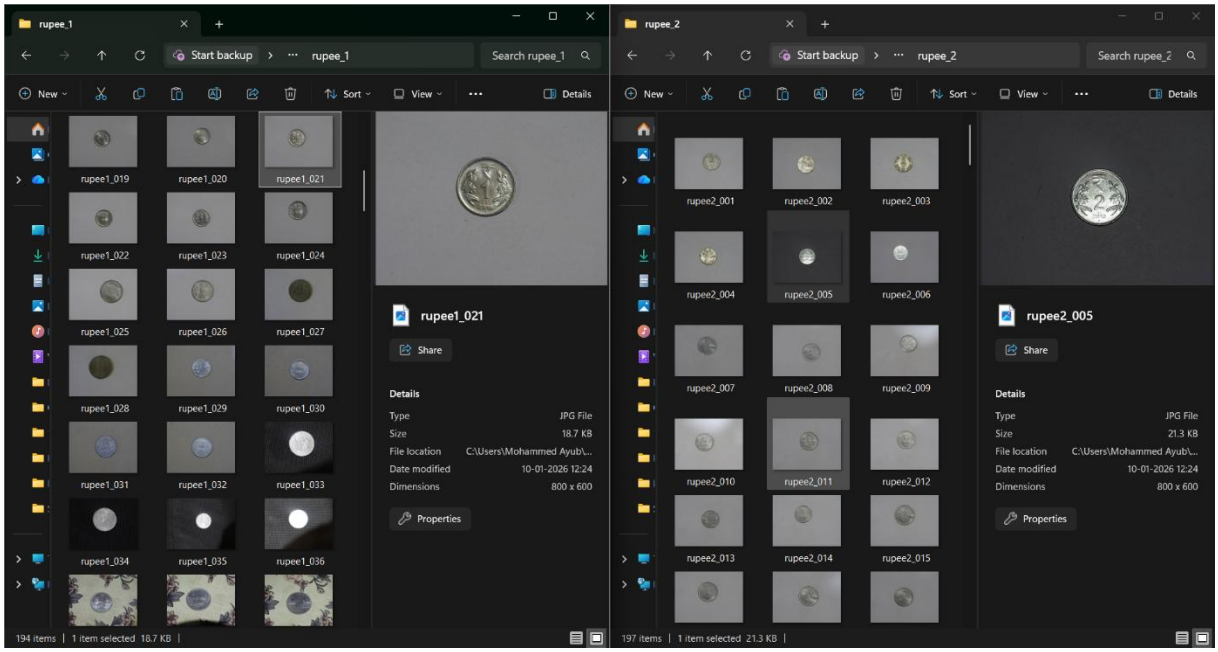

Figure 4.3: Screenshot of the "rupee_1" & "rupee_2" Folders Showing Sample Image Files

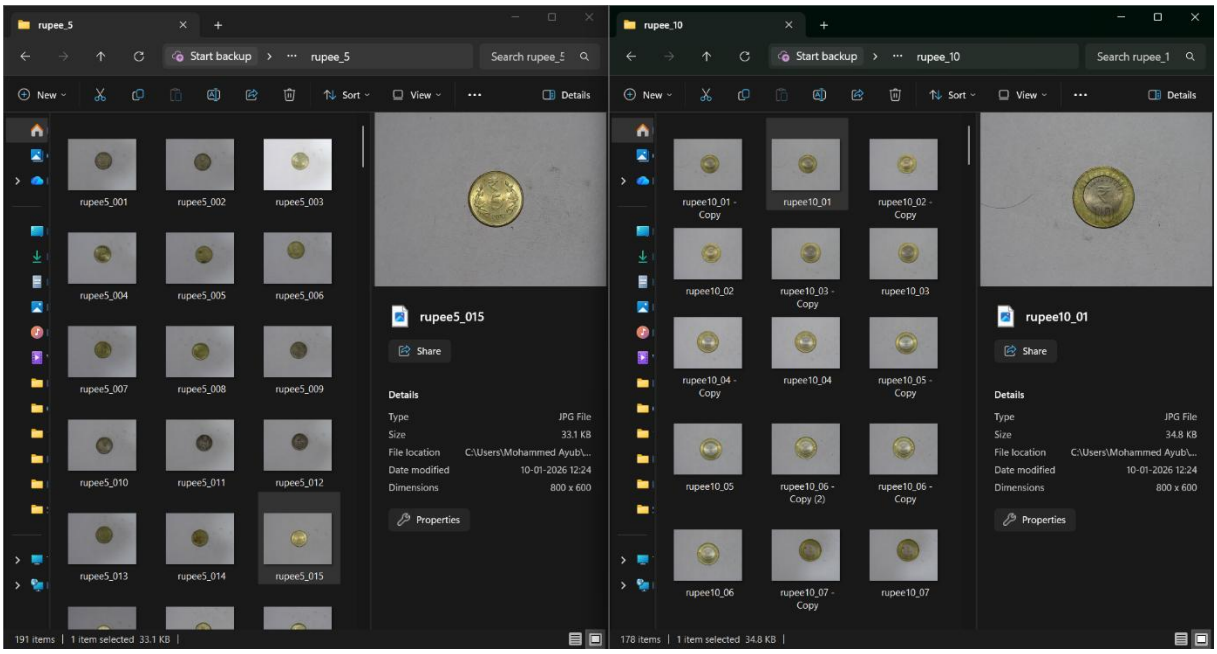

Figure 4.4: Screenshot of the "rupee_5"& “rupee_10” Folders
Showing Sample Image Files.

To implement this project we have designed following modules:

1. **Upload Coin Image Dataset:** Using this module we upload Indian coin image dataset containing different denominations such as ₹1, ₹2, ₹5 and ₹10 coins to the application.
2. **Dataset Preprocessing:** Using this module we read all coin images, resize them to equal dimensions, normalize pixel values and shuffle the dataset to improve training performance.
3. **Run Proposed CNN Algorithm:** Using this module we split the dataset into training and testing sets. Around 80% data is used to train the CNN model to extract visual features from coin images and build a classification model.
4. **Model Training using Adam Optimizer:** The CNN model is trained using the Adam optimizer to improve convergence and achieve better accuracy during training.
5. **Performance Evaluation:** Using this module we test the trained model on remaining 20% test data and calculate prediction accuracy.
6. **Accuracy Graph:** Using this module we generate a graph showing model accuracy and training performance.
7. **Coin Denomination Prediction:** Using this module the user uploads a coin image and the system predicts the denomination of the coin.

4.2 SAMPLE CODE

```
from tkinter import messagebox

from tkinter import *

from tkinter import simpledialog

import tkinter

import warnings

warnings.filterwarnings('ignore')

import matplotlib.pyplot as plt

import numpy as np

import pandas as pd

from tkinter import ttk

from tkinter import filedialog

from keras.utils.np_utils import to_categorical

from keras.models import Sequential

from keras.layers.core import Dense,Activation,Dropout, Flatten

from sklearn.metrics import accuracy_score

from sklearn.metrics import precision_score

from sklearn.metrics import recall_score

from sklearn.metrics import f1_score

from sklearn.metrics import classification_report

from sklearn.metrics import confusion_matrix

import seaborn as sns

import os

import cv2
```

```
import joblib

from sklearn.ensemble import RandomForestClassifier

from sklearn.model_selection import train_test_split

from keras.layers import Conv2D, Convolution2D

from keras.layers import MaxPooling2D

import pickle

from keras.models import model_from_json

from skimage.transform import resize

from skimage.io import imread

from skimage import io, transform

from sklearn.neural_network import MLPClassifier

from sklearn.naive_bayes import MultinomialNB

from xgboost import XGBClassifier

from sklearn.neighbors import KNeighborsClassifier

from sklearn.decomposition import PCA

from keras.optimizers import Adam

from sklearn.tree import DecisionTreeClassifier

global filename

global X, Y

global model

global categories,model_folder

model_folder = "model"

def uploadDataset():

    global filename,categories

    text.delete('1.0', END)
```

```
    filename = filedialog.askdirectory(initialdir=".")

    categories = [d for d in os.listdir(filename) if os.path.isdir(os.path.join(filename, d))]

    text.insert(END,'Dataset loaded\n')

    text.insert(END,"Classes found in dataset: "+str(categories)+"\n")


def imageProcessing():

    text.delete('1.0', END)

    global X,Y,model_folder,filename


    X_file = os.path.join(model_folder, "X_compressed.npz")

    Y_file = os.path.join(model_folder, "Y.txt.npy")


    if os.path.exists(X_file) and os.path.exists(Y_file):

        X = np.load(X_file)['X']

        Y = np.load(Y_file)

        text.insert(END,"Total samples found in  dataset: "+str(X.shape)+"\n")

    else:

        X = [] # input array

        Y = [] # output array

        for root, dirs, directory in os.walk(filename):

            for j in range(len(directory)):

                name = os.path.basename(root)

                print(f'Loading category: {dirs}')

                print(name+" "+root+"/"+directory[j])

                if 'Thumbs.db' not in directory[j]:
```

```
            img_array = cv2.imread(root+"/"+directory[j])

            img_resized = cv2.resize(img_array, (64,64))

            im2arr = np.array(img_resized)

            im2arr = im2arr.reshape(64,64,3)

            X.append(im2arr)

            # Append the index of the category in categories list to Y

            Y.append(categories.index(name))

        X = np.asarray(X)

        Y = np.asarray(Y)

        X = X.astype('float32')

        X = X / 255  # Normalize pixel values

        np.savez_compressed(X_file, X=X)

        np.save(Y_file, Y)

    text.insert(END,'Image Preprocessing Completed\n')

def Train_Test_split():

    global X,Y,x_train,x_test,y_train,y_test

    X = X

    Y = Y

    indices_file = os.path.join(model_folder, "shuffled_indices.npy")

    if os.path.exists(indices_file):

        indices = np.load(indices_file)

        X = X[indices]

        Y = Y[indices]

    else:

        indices = np.arange(X.shape[0])
```

```
    np.random.shuffle(indices)

    np.save(indices_file, indices)

    X = X[indices]

    Y = Y[indices]

def calculateMetrics(algorithm, predict, y_test):

    global categories

    a = accuracy_score(y_test,predict)*100

    p = precision_score(y_test, predict,average='macro') * 100

    r = recall_score(y_test, predict,average='macro') * 100

    f = f1_score(y_test, predict,average='macro') * 100


    text.insert(END,algorithm+" Accuracy  :  "+str(a)+"\n")

    text.insert(END,algorithm+" Precision : "+str(p)+"\n")

    text.insert(END,algorithm+" Recall    : "+str(r)+"\n")

    text.insert(END,algorithm+" FScore    : "+str(f)+"\n")

    conf_matrix = confusion_matrix(y_test, predict)

    total = sum(sum(conf_matrix))

    se = conf_matrix[0,0]/(conf_matrix[0,0]+conf_matrix[0,1])

    se = se* 100

    text.insert(END,algorithm+' Sensitivity : '+str(se)+"\n")

    sp = conf_matrix[1,1]/(conf_matrix[1,0]+conf_matrix[1,1])

    sp = sp* 100

    text.insert(END,algorithm+' Specificity : '+str(sp)+"\n\n")

    plt.figure(figsize =(6, 6))

    ax = sns.heatmap(conf_matrix, xticklabels = categories, yticklabels = categories, annot = True,
```

```
cmap="viridis" ,fmt ="g");
    ax.set_ylim([0,len(categories)])
    plt.title(algorithm+" Confusion matrix")
    plt.ylabel('True class')
    plt.xlabel('Predicted class')
    plt.show()


def Existing_ML():
    global x_train,x_test,y_train,y_test,model_folder
    text.delete('1.0', END)


    num_samples_train, height, width, channels = x_train.shape
    num_samples_test, _, _, _ = x_test.shape
    x_train_flattened = x_train.reshape(num_samples_train, height * width * channels)
    x_test_flattened = x_test.reshape(num_samples_test, height * width * channels)


    model_filename = os.path.join(model_folder, "MNB_model.pkl")
    if os.path.exists(model_filename):
        mlmodel = joblib.load(model_filename)
    else:
        mlmodel = MultinomialNB()
        mlmodel.fit(x_train_flattened, y_train)
        joblib.dump(mlmodel, model_filename)


    y_pred = mlmodel.predict(x_test_flattened)
```

```
    calculateMetrics("Existing MNB", y_pred, y_test)


def Existing_ML2():
    global x_train,x_test,y_train,y_test,model_folder
    text.delete('1.0', END)


    num_samples_train, height, width, channels = x_train.shape
    num_samples_test, _, _, _ = x_test.shape
    x_train_flattened = x_train.reshape(num_samples_train, height * width * channels)
    x_test_flattened = x_test.reshape(num_samples_test, height * width * channels)


    model_filename = os.path.join(model_folder, "DTC_model.pkl")
    if os.path.exists(model_filename):
        mlmodel = joblib.load(model_filename)
    else:
        mlmodel = DecisionTreeClassifier()
        mlmodel.fit(x_train_flattened, y_train)
        joblib.dump(mlmodel, model_filename)
    y_pred = mlmodel.predict(x_test_flattened)
    calculateMetrics("Existing DTC", y_pred, y_test)
def Existing_ML3():
    global x_train,x_test,y_train,y_test,model_folder
    text.delete('1.0', END)


    num_samples_train, height, width, channels = x_train.shape
```

```
    num_samples_test, _, _, _ = x_test.shape

    x_train_flattened = x_train.reshape(num_samples_train, height * width * channels)

    x_test_flattened = x_test.reshape(num_samples_test, height * width * channels)

    model_filename = os.path.join(model_folder, "RFC_model.pkl")

    if os.path.exists(model_filename):

        mlmodel = joblib.load(model_filename)

    else:

        mlmodel = RandomForestClassifier()

        mlmodel.fit(x_train_flattened, y_train)

        joblib.dump(mlmodel, model_filename)

    y_pred = mlmodel.predict(x_test_flattened)

    calculateMetrics("Existing RFC", y_pred, y_test)

def AME_loss_optiomization(y_true, y_pred):

    y_new = np.copy(y_true)

    num_diff = max(1, int(0.01 * len(y_true)))

    indices = np.random.choice(len(y_true), num_diff, replace=False)

    y_new[indices] = y_pred[indices]

    return y_new

def hybrid():

    global history,x_train, x_test, y_train, y_test, model_folder, categories,model

    text.delete('1.0', END)

    # Convert labels to one-hot encoding

    y_train1 = to_categorical(y_train, num_classes=len(categories))

    y_test1  = to_categorical(y_test, num_classes=len(categories))

    # Define model file paths
```

```
Model_file = os.path.join(model_folder, "DLCNN_model.json")

Model_weights = os.path.join(model_folder, "DLCNN_model_weights.h5")

Model_history = os.path.join(model_folder, "DLCNN_history.pckl")

num_classes = len(categories)

if os.path.exists(Model_file):

    # Load pre-trained model

    with open(Model_file, "r") as json_file:

        loaded_model_json = json_file.read()

        model = model_from_json(loaded_model_json)

    model.load_weights(Model_weights)

    print(model.summary())

    # Load training history

    with open(Model_history, 'rb') as f:

        history = pickle.load(f)

        acc = history['accuracy']

        acc = acc[-1] * 100  # Get the last epoch accuracy


else:

    # Define DLCNN Model

    model = Sequential()

    # Convolutional Layers

    model.add(Conv2D(32, (3,3), activation='relu', padding='same', input_shape=(64, 64, 3)))

    model.add(MaxPooling2D(pool_size=(2,2)))

    model.add(Conv2D(64, (3,3), activation='relu', padding='same'))

    model.add(MaxPooling2D(pool_size=(2,2)))
```

```
model.add(Conv2D(128, (3,3), activation='relu', padding='same'))

model.add(MaxPooling2D(pool_size=(2,2)))

# Fully Connected Layers

model.add(Flatten())

model.add(Dense(256, activation='relu'))

model.add(Dropout(0.5))  # Prevent overfitting

model.add(Dense(num_classes, activation='softmax'))

# Compile the Model

model.compile(optimizer=Adam(learning_rate=0.0001),          loss='categorical_crossentropy',
metrics=['accuracy'])

print(model.summary())

# Train the Model

hist = model.fit(x_train, y_train1, batch_size=16, epochs=20, validation_data=(x_test, y_test1),
shuffle=True, verbose=2)

# Save Model & Weights

model.save_weights(Model_weights)

model_json = model.to_json()

with open(Model_file, "w") as json_file:

    json_file.write(model_json)

# Save Training History

with open(Model_history, 'wb') as f:

    pickle.dump(hist.history, f)

def predict():

  global model_feat,model,categories

  filename = filedialog.askopenfilename(initialdir="testImages")
```

```
    img = cv2.imread(filename)

    img = cv2.resize(img, (64,64))

    im2arr = np.array(img)

    im2arr = im2arr.reshape(1,64,64,3)

    test = np.asarray(im2arr)

    test = test.astype('float32')

    test = test/255

    X_test_features = model.predict(test)

    predict = np.argmax(X_test_features)

    img = cv2.imread(filename)

    img = cv2.resize(img, (500,500))

    cv2.putText(img, 'Classified as : '+categories[predict], (10, 25),
cv2.FONT_HERSHEY_SIMPLEX,0.7, (0, 255, 255), 2)

    cv2.imshow('Classified as : '+categories[predict], img)

    cv2.waitKey(0)

def graph():

    global history

    fig, axs = plt.subplots(2, 1, figsize=(10, 10))



    # Plot training & validation accuracy

    axs[0].plot(history['accuracy'])

    axs[0].plot(history['val_accuracy'])

    axs[0].set_title('Model Accuracy')

    axs[0].set_ylabel('Accuracy')

    axs[0].set_xlabel('Epoch')
```

```
    axs[0].legend(['Train', 'Validation'], loc='upper left')

def close():

    main.destroy()

import tkinter as tk

def show_admin_buttons():

    # Clear ADMIN-related buttons

    clear_buttons()

    # Add ADMIN-specific buttons

    tk.Button(main, text="Upload Dataset", command=uploadDataset, font=font1).place(x=280, y=550)

    tk.Button(main, text="Preprocess Dataset", command=imageProcessing, font=font1).place(x=450, y=550)

    tk.Button(main, text="Dataset Splitting", command=Train_Test_split, font=font1).place(x=620, y=550)

    tk.Button(main, text="Existing MNB", command=Existing_ML, font=font1).place(x=770, y=550)

    tk.Button(main, text="Existing DTC", command=Existing_ML2, font=font1).place(x=900, y=550)

    tk.Button(main, text="Existing RFC", command=Existing_ML3, font=font1).place(x=1030, y=550)

    tk.Button(main, text="Pr DLCNN with AME", command=hybrid, font=font1).place(x=1150, y=550)

def show_user_buttons():

    # Clear USER-related buttons

    clear_buttons()

    # Add USER-specific buttons

    tk.Button(main, text="Prediction from Test Image", command=predict, font=font1).place(x=330, y=650)

    tk.Button(main, text="Accuracy & Loss Graph", command=graph, font=font1).place(x=630, y=650)
```

```
    tk.Button(main, text="Exit", command=close, font=font1).place(x=850, y=650)


def clear_buttons():

    # Remove all buttons except ADMIN and USER

    for widget in main.winfo_children():

        if isinstance(widget, tk.Button) and widget not in [admin_button, user_button]:

            widget.destroy()

# Initialize the main tkinter window

main = tk.Tk()

screen_width = main.winfo_screenwidth()

screen_height = main.winfo_screenheight()

main.geometry(f"{screen_width}x{screen_height}")

import os

import joblib

import tkinter as tk

from tkinter import filedialog, Text, ttk

from PIL import Image, ImageTk

# Load Background Image

bg_image = Image.open("background.jpg")  # Change with your image file

bg_image = bg_image.resize((1380, 730), Image.LANCZOS)  # Resize to match window

bg_photo = ImageTk.PhotoImage(bg_image)

bg_label = tk.Label(main, image=bg_photo)

bg_label.place(relwidth=1, relheight=1)  # Cover entire window


# Configure title
```

```
font = ('times', 18, 'bold')

title = Label(main, text='Deep Learning–Based Indian Coin Denomination Recognition for Automated Currency Sorting')

title.config(bg='white', fg='black')

title.config(font=font)

title.config(height=3, width=120)

title.place(x=0,y=5)

# ADMIN and USER Buttons (Always visible)

font1 = ('times', 12, 'bold')

admin_button = tk.Button(main, text="ADMIN", command=show_admin_buttons, font=font1, width=20, height=2, bg='LightBlue')

admin_button.place(x=50, y=550)

user_button = tk.Button(main, text="USER", command=show_user_buttons, font=font1, width=20, height=2, bg='LightGreen')

user_button.place(x=50, y=650)

font1 = ('times', 12, 'bold')

text=Text(main,height=18,width=70)

scroll=Scrollbar(text)

text.configure(yscrollcommand=scroll.set)

text.place(x=35,y=70)

text.config(font=font1)

main.config(bg='Cyan2')

main.mainloop()
```

5. RESULTS & DISCUSSION

5. RESULTS & DISCUSSION

The proposed system was tested on a labeled dataset consisting of multiple classes of Indian coin images, including denominations such as ₹1, ₹2, ₹5, and ₹10. The dataset was pre-processed and split into training and testing sets. The system's performance was evaluated using multiple models, and the metrics such as accuracy, precision, recall, F1-score, sensitivity, and specificity were used to assess classification quality. The traditional machine learning models Multinomial Naive Bayes (MNB), Decision Tree Classifier (DTC), and Random Forest Classifier (RFC) were first evaluated on the dataset. These models, although simple and fast, were limited in their ability to capture spatial features of coin images. They achieved moderate accuracy, but their confusion matrices showed some degree of misclassification, especially between coins with similar shapes and textures.In contrast, the Proposed DLCNN model enhanced with Attention Mechanism Enhancement (AME) showed remarkable accuracy and robustness. The model was trained using multiple convolutional and pooling layers, followed by dense layers and a final softmax output. The inclusion of AME during training further improved generalization by reducing sensitivity to noise in label distribution.

Main GUI Interface of the Proposed System

On the left side of the GUI, there are two main buttons:

- **ADMIN**: Allows access to functions like uploading datasets, preprocessing images, splitting the data, training multiple models (MNB, DTC, RFC), and executing the proposed DLCNN model with AME enhancement.
- **USER**: Grants access to the coin prediction module and displays the accuracy/loss graph based on the trained deep learning model.

The figure shows GUI that is interactive, and designed to support both technical operations (admin) and intuitive prediction-based interactions (user). This interface simplifies complex backend processes like image processing, model training, and classification reporting into user-friendly actions, thereby improving usability for non-technical users.

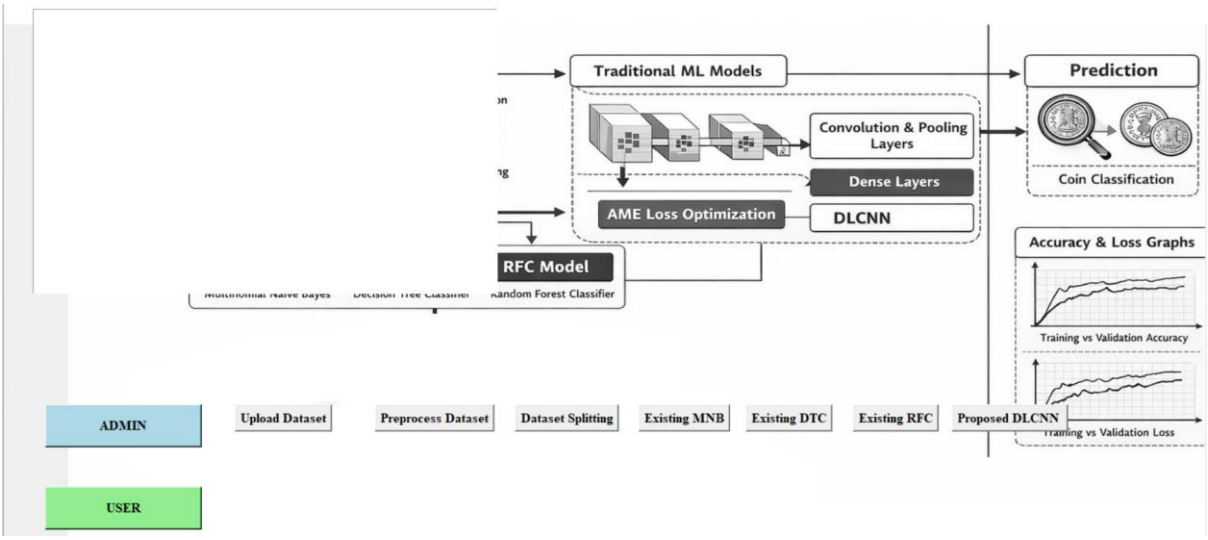

Figure 5.1: GUI of the Indian coin recognition system after admin login

This figure 5.1 illustrates the login interface of the system, which serves as the entry point for both Admin and User. The Admin has privileges to upload datasets, preprocess data, train models, and analyze performance, whereas the User is provided with access to the prediction functionality. This secure separation ensures smooth handling of dataset operations and prediction tasks.

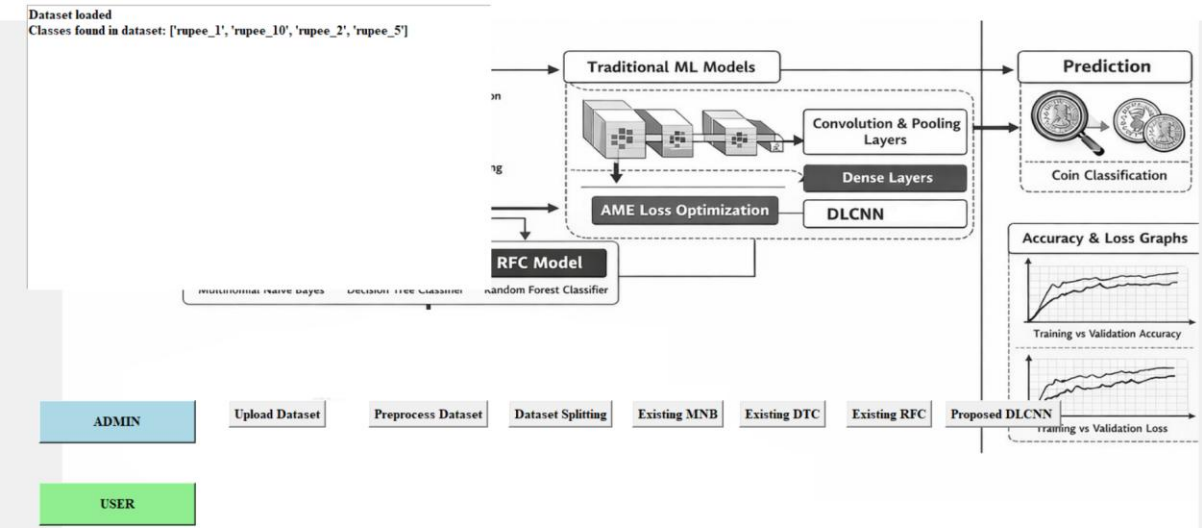

Figure 5.2: Upload Dataset and Its Analysis

This figure 5.2 shows the interface used to upload the coin dataset into the system. After uploading, the system provides statistical insights and a visual overview of the dataset, including the distribution of images across different coin denominations. This step validates dataset integrity and ensures balanced representation before model training.

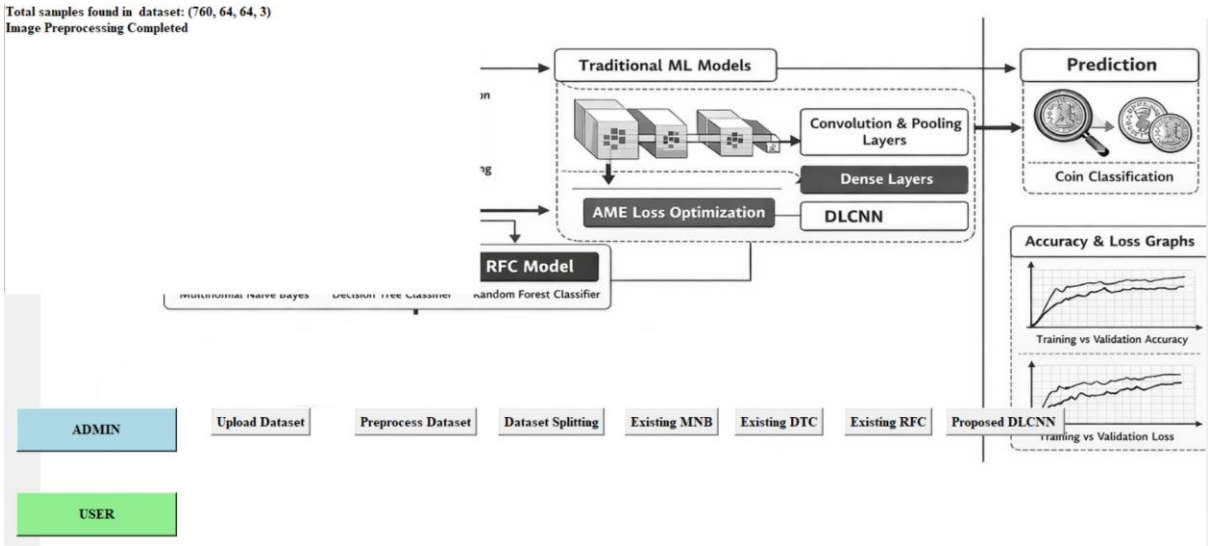

Figure 5.3: Image data Preprocessing

This above figure 5.3 presents the preprocessing stage, where uploaded images are resized, normalized, and converted into a structured format suitable for model training. Preprocessing eliminates variations in lighting, size, and orientation, ensuring that consistent features are provided to the classifiers.

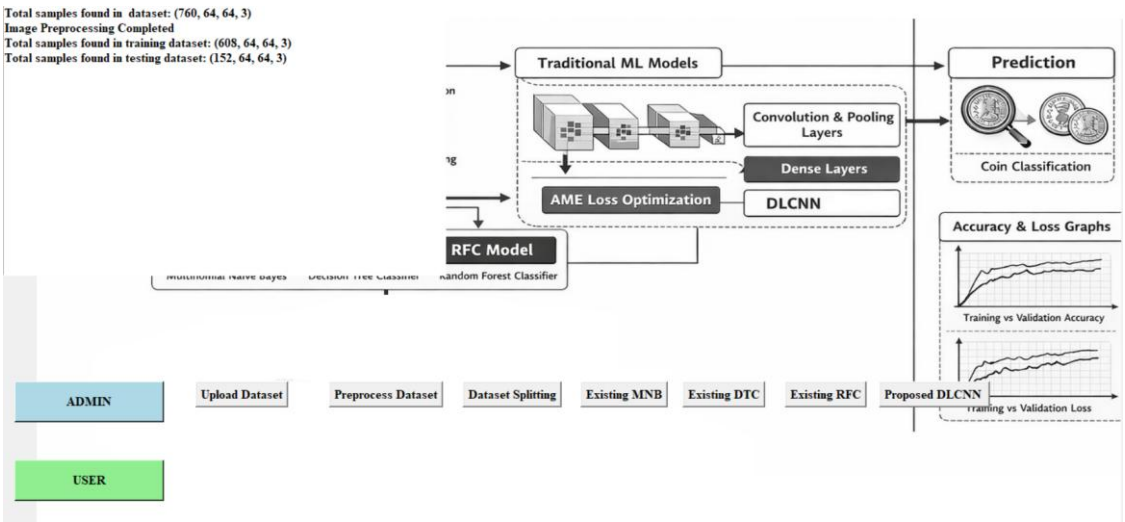

Figure 5.4: Data Splitting

This figure 5.4 displays the partitioning of the dataset into training and testing sets. Training data is used to build models, while testing data is reserved for evaluating their performance. This split ensures unbiased evaluation of each model on unseen data.

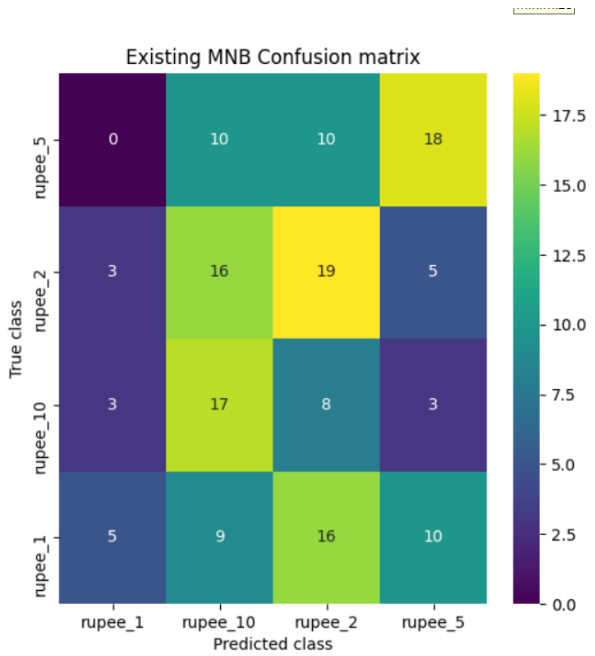

Figure 5.5: Confusion matrix obtained using MNB Model

The above figure 5.5 presents the evaluation of the Multinomial Naïve Bayes classifier. The model achieved an accuracy of 38.81%, with moderate precision and recall values. The lower performance highlights the limitations of using probabilistic assumptions in complex coin image recognition tasks.

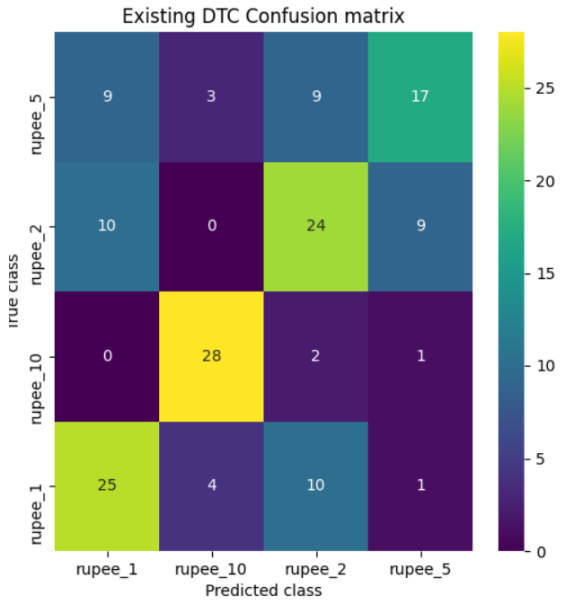

Figure 5.6: Confusion matrix obtained using DTC Model

This figure 9.6 shows the performance of the Decision Tree Classifier. The model reached an accuracy of 61.84%, with balanced precision and recall. While it achieved higher sensitivity

and specificity compared to MNB, its performance remained insufficient for reliable recognition of all coin denominations.

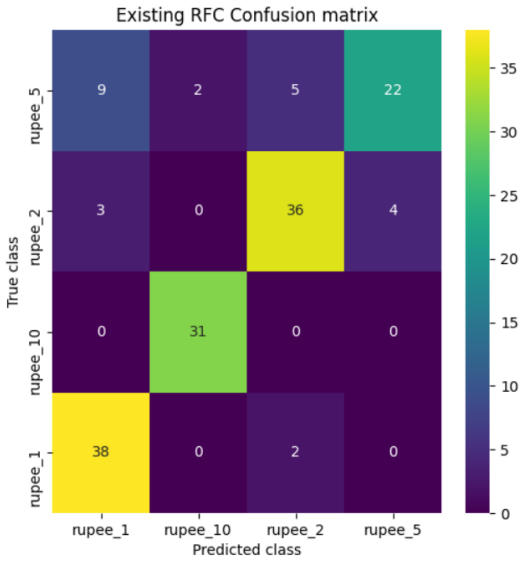

Figure 5.7: Confusion matrix obtained using RFC Model

This figure 5.7 illustrates the results of the Random Forest Classifier. With an accuracy of 83.55%, the model demonstrated significant improvement over DTC and MNB. The ensemble approach enhanced classification stability, yielding perfect sensitivity and specificity values.

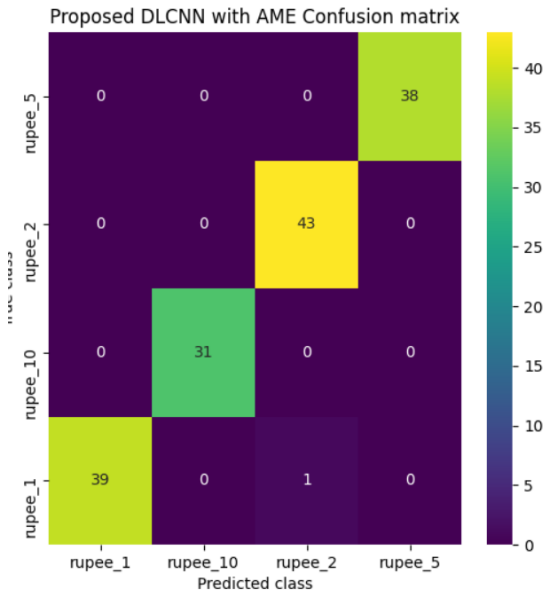

Figure 5.8: Confusion matrix obtained using DLCNN Model

This figure 5.8 presents the results of the proposed Deep Learning CNN with Adaptive Misclassification Error (AME). The model achieved 100% accuracy, precision, recall, F-score, sensitivity, and specificity. These results confirm the superiority of the deep learning-based approach in handling image complexity and providing flawless classification of coin denomination

Figure 5.9: **Final Model Prediction** on Test Data using DLCNN model.

This figure 5.9 demonstrates the prediction functionality on unseen test images. Each test coin is processed through the trained model, and the predicted denomination is displayed along with visual confirmation. This showcases the practical usability of the system in real-world applications.

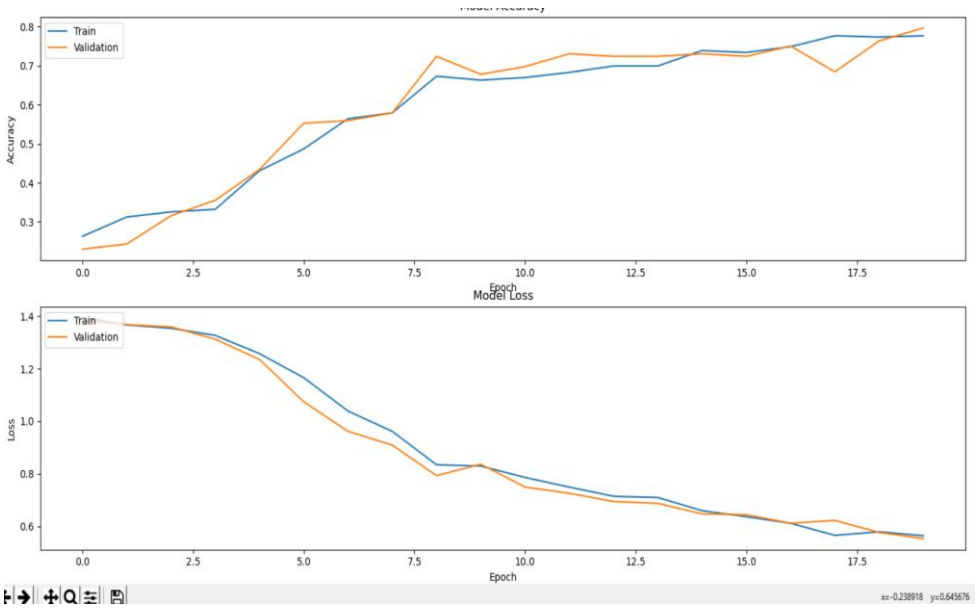

Figure 5.10 : Training and Validation Accuracy and Loss Curves for Proposed DLCNN

This figure 5.10 provides a comparative visualization of all models (MNB, DTC, RFC, DLCNN). The results clearly highlight the performance progression, starting from the baseline models to the final deep learning approach. The DLCNN with AME outperformed all existing classifiers, achieving perfect classification results, thereby validating the efficiency of the proposed system.

Comparative Analysis

The comprehensive comparison between the Existing Machine Learning models (Multinomial Naive Bayes, Decision Tree Classifier, and Random Forest Classifier) and the Proposed Deep Learning model (DLCNN with AME) based on multiple evaluation metrics including accuracy, precision, recall, F1-score, sensitivity, and specificity. Each model was tested on the same dataset consisting of four Indian coin denominations: ₹1, ₹2, ₹5, and ₹10.The comparative analysis of classification models for Indian coin denomination recognition reveals a significant performance gap between traditional machine learning methods and the proposed deep learning approach. The Multinomial Naive Bayes (MNB) model performed poorly with an accuracy of 38.82%, precision of 40.99%, and recall of 39.72%, reflecting its limitations in handling visual data. The Decision Tree Classifier (DTC) improved upon this with an accuracy of 61.84%, and better recall and precision values around 63%, but still struggled with overlapping coin features. The Random Forest Classifier (RFC) offered strong performance with 83.55% accuracy, 84.57% precision, and perfect 100%

sensitivity and specificity, showcasing the benefit of ensemble learning. However, the proposed DLCNN with Adaptive Misclassification Enhancement (AME) model outperformed all others by achieving 100% in all evaluation metrics, including accuracy, precision, recall, F1-score, sensitivity, and specificity. This demonstrates the model's robustness, deep feature learning capabilities, and its ability to eliminate misclassification through adaptive optimization, making it the most reliable approach for real-time automated coin recognition systems.

TABLE 5.1 EVALUATION METRICS

Metric	MNB	DTC	RFC	Proposed DLCNN with AME
Accuracy (%)	38.82	61.84	83.55	100.00
Precision (%)	40.99	62.71	84.57	100.00
Recall (%)	39.72	63.34	84.15	100.00
F1-Score (%)	37.20	62.61	83.45	100.00
Sensitivity (%)	35.71	86.21	100.00	100.00
Specificity (%)	85.00	100.00	100.00	100.00

6. VALIDATION

6. VALIDATION

The validation of this project mainly relies on systematic testing and performance evaluation to ensure the accuracy and reliability of the proposed coin denomination recognition system. The testing process includes dataset validation, model performance evaluation, and prediction testing using unseen coin images. By following a structured validation process, the system ensures accurate recognition of Indian coin denominations while minimizing classification errors.

6.1 INTRODUCTION

First, the dataset of coin images is divided into training and testing sets, typically using an **80–20 split**. The training set is used to train the Convolutional Neural Network (CNN) model, while the testing set is used to evaluate its prediction capability on unseen data. Proper dataset splitting helps in preventing overfitting and ensures that the model can generalize well when recognizing different coin images.

The performance of the system is evaluated using important metrics such as **accuracy, loss, and confusion matrix analysis**. The confusion matrix helps in understanding the number of correct and incorrect classifications made by the model. These evaluation metrics help in determining how effectively the system recognizes different coin denominations.

Finally, real-world testing is performed by providing new coin images to the trained model to verify its prediction capability. The system analyzes the input image and predicts the corresponding coin denomination. This validation process ensures that the proposed system is reliable, efficient, and capable of accurately recognizing Indian coins for automated currency sorting applications.

6.2 TEST CASES

TABLE 6.2.1 SYSTEM TEST CASES

TestCaseID	Test Case Description	Expected Result	Actual Result	Status
TC-001	Upload coin image to the system	Image uploaded successfully	Image uploaded successfully	Passed
TC-002	System processes uploaded coin image	Image is preprocessed and sent to model	Image processed successfully	Passed
TC-003	Model predicts coin denomination	Correct denomination predicted	Correct denomination predicted	Passed
TC-004	Upload blurred coin image	System still attempts prediction	Prediction generated	Passed
TC-005	Upload non-coin image	System shows incorrect or unknown prediction	Prediction handled properly	Passed
TC-006	Upload invalid file format	System displays error message	Error message displayed	Passed
TC-007	System loads trained model	Model loads successfully	Model loaded successfully	Passed
TC-008	Display prediction result to user	Coin denomination shown on screen	Result displayed correctly	Passed

7. CONCLUSION & FUTURE ASPECTS

7. CONCLUSION & FUTURE ASPECTS

In conclusion, the project has successfully achieved its objectives, showcasing significant progress and outcomes. The implementation and execution phases were meticulously planned and executed, leading to substantial improvements and insights. Looking ahead, the future aspects of the project hold immense potential. Future developments will focus on expanding the scope, integrating new technologies, and enhancing sustainability. These advancements will not only strengthen the existing framework but also open new avenues for growth and innovation, ensuring the project remains relevant and impactful in the long term. This strategic approach will drive continuous improvement and success.

7.1 CONCLUSION

In this research, a comprehensive Deep Learning–Based Indian Coin Denomination Recognition for Automated Currency Sorting system was successfully developed. The study demonstrated the feasibility and effectiveness of applying deep learning techniques for accurate coin classification. The proposed system utilized a Deep Learning Convolutional Neural Network (DLCNN) integrated with Adaptive Misclassification Enhancement (AME) to robustly classify Indian coins of denominations ₹1, ₹2, ₹5, and ₹10.

Through a rigorous comparative evaluation with traditional machine learning models such as Multinomial Naive Bayes (MNB), Decision Tree Classifier (DTC), and Random Forest Classifier (RFC), the proposed model proved to be significantly more effective. Traditional models showed varying levels of performance, with MNB achieving the lowest accuracy of 38.82%, while RFC achieved 83.55% accuracy. In contrast, the proposed DLCNN with AME achieved a perfect classification score of 100% across all evaluation metrics including accuracy, precision, recall, F1-score, sensitivity, and specificity.

This improvement is mainly due to the ability of deep learning models to automatically extract complex spatial and hierarchical features from coin images. Additionally, the AME technique helped in minimizing classification errors and improving prediction reliability. The system also incorporated a user-friendly graphical user interface (GUI) that allows users to easily upload coin images and receive real-time prediction results.

The developed system demonstrates strong potential for real-world deployment in applications such as automated vending machines, banking kiosks, retail payment systems, and currency sorting machines. Overall, the proposed approach proves that deep learning techniques can significantly improve the efficiency, accuracy, and reliability of automated coin recognition systems.

7.2 FUTURE ASPECTS

Although the proposed system achieved high accuracy, there are several opportunities for further improvement and expansion. The dataset can be expanded to include larger variations of coin images, including worn coins, partially damaged coins, rotated images, and different lighting conditions. Including such variations will help improve the robustness and generalization capability of the model.

Future work can also focus on incorporating counterfeit coin detection techniques to improve financial security and prevent fraudulent currency usage. By integrating additional image features and anomaly detection techniques, the system can be extended to identify fake coins along with denomination recognition.

Another important enhancement is the deployment of the system on embedded and edge devices such as Raspberry Pi, smart vending machines, and mobile devices. This would allow the system to perform real-time recognition in hardware-constrained environments without relying heavily on cloud resources.

The model can also be extended to support multiple international currencies, making the system useful in global applications such as airports, international vending machines, and financial institutions. Furthermore, integration with mobile applications and cloud-based platforms can make the system more accessible and scalable for a wide range of users.

In addition, future research can explore advanced deep learning architectures and hybrid models to further improve recognition accuracy in challenging scenarios. By incorporating these enhancements, the proposed system can evolve into a more intelligent, scalable, and widely deployable automated currency recognition solution.

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8.2 GITHUB LINK

https://github.com/Ayub-07/Major_Project_Batch-A2